MMP 1
Serie 4
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1 MC

(A)/(B)

2

2.a

$$f(x) = e^{-|x|} \tag{2.1}$$

$$\begin{aligned}
\hat{f}(\omega) &= \int_{-\infty}^{\infty} e^{-|x|} e^{-i\omega x} dx \\
&= \left[\int_{0}^{\infty} e^{-x(i\omega+1)} dx + \int_{-\infty}^{0} e^{-x(i\omega-1)} dx \right] \\
&= \left[\frac{-1}{i\omega+1} e^{-x(i\omega+1)} \right]_{0}^{\infty} + \left[\frac{-1}{i\omega-1} e^{-x(i\omega-1)} \right]_{-\infty}^{0} \\
&= \left(\frac{1}{i\omega+1} + \frac{-1}{i\omega-1} \right) \\
&= \frac{2}{1+\omega^2}
\end{aligned} \tag{2.2}$$

2.b

$$\sum_{n\in\mathbb{Z}} f(na) = \frac{1}{a} \sum_{n\in\mathbb{Z}} \hat{f}\left(\frac{2\pi n}{a}\right) \tag{2.3}$$

$a = 2\pi$

$$\sum_{n\in\mathbb{Z}} e^{-|n2\pi|} = \frac{1}{2\pi} \sum_{n\in\mathbb{Z}} 2\frac{1}{1+n^2} \tag{2.4}$$

$$\Rightarrow 2\sum_{n=1}^{\infty} \frac{1}{1+n^2} + 1 = \pi \left(2\sum_{n=1}^{\infty} e^{-n2\pi} + 1 \right) \tag{2.5}$$

$$\begin{aligned}
\Rightarrow \sum_{n=0}^{\infty} \frac{1}{1+n^2} &= \frac{\pi}{2} \left(2\sum_{n=0}^{\infty} e^{-n2\pi} - 1 \right) + \frac{1}{2} \\
&= \pi \sum_{n=0}^{\infty} e^{-n2\pi} + \frac{1-\pi}{2} \\
&= \frac{\pi}{1-e^{-2\pi}} + \frac{1-\pi}{2} \\
&= -\frac{\pi}{2}\left(1 + \frac{2}{e^{-2\pi}-1} - 1\right) + \frac{1-\pi}{2} \\
&= \frac{\pi}{2}\coth(\pi) + \frac{1}{2}
\end{aligned} \tag{2.6}$$

3

$$f(x) = \frac{1}{2} - x, \quad \forall x \in [0,1) \tag{3.1}$$

$$\begin{aligned}
f_n &= \int_{0}^{1} \left(\frac{1}{2} - x\right) e^{-2\pi i n x} dx \\
&= \left[\frac{1}{-2\pi i n} e^{-2\pi i n x} + \frac{x}{2\pi i n} e^{-2\pi i n x} \right]_{0}^{1} - \underbrace{\int_{0}^{1} \ldots}_{=0} \\
&= \frac{1}{2\pi i n}
\end{aligned} \tag{3.2}$$

$n = 0$

$$
\begin{aligned}
f_0 &= \int \frac{1}{2} - x dx \\
&= \left[\frac{1}{2}x - \frac{1}{2}x^2\right]_0^1 = 0
\end{aligned} \tag{3.3}
$$

3.a

$$
\begin{aligned}
s_N f(x) &= \sum_{0<|n|\leq N} \frac{1}{2\pi i n} e^{2\pi n i x} \\
&= \sum_{n=1}^{N} \frac{1}{\pi n} \sin(2\pi n x)
\end{aligned} \tag{3.4}
$$

$$
\begin{aligned}
\frac{d}{dx}\left(s_n f(x) - \frac{1}{2} + x\right) &= 1 + \sum_{n=1}^{N} \frac{2\pi n}{\pi n} \cos(2\pi n x) \\
&= 1 + 2\sum_{n=1}^{N} \cos(2\pi n x) \\
&= \frac{\sin\left(\left(N+\frac{1}{2}\right)2\pi x\right)}{\sin\left(\frac{2\pi}{2}x\right)} = 0
\end{aligned} \tag{3.5}
$$

$$
\Rightarrow 1 = (2N+1)x \Leftrightarrow x = \frac{1}{2N+1} \tag{3.6}
$$

3.b

$$
E'_N = \left(\frac{\sin((2N+1)\pi x)}{\sin((\pi x))}\right) \tag{3.7}
$$

$$
\begin{aligned}
\Rightarrow E_N(x_{\max}) &= E(0) + \int_0^{x_{\max}} \frac{\sin((2N+1)\pi x)}{\sin((\pi x))} dx \\
&= -\frac{1}{2} + \int_0^{\frac{1}{2N+1}} \frac{\sin((2N+1)\pi x)}{\sin((\pi x))} dx
\end{aligned} \tag{3.8}
$$

3.c

$$
\begin{aligned}
E_N(x_{\max}) &= -\frac{1}{2} + \int_0^{\frac{1}{2N+1}} \frac{\sin((2N+1)\pi x)}{\sin((\pi x))} dx \\
&\approx -\frac{1}{2} + \int_0^{\frac{1}{2N+1}} \frac{\sin((2N+1)\pi x)}{\pi x + O(x^2)} dx \\
&\approx -\frac{1}{2} + \frac{1}{\pi}\int_0^{\pi} \frac{\sin(x)}{x} dx \\
&= -\frac{1}{2} + \frac{1}{\pi}\sum_{n=0}^{\infty} (-1)^n \frac{\pi^{2n+1}}{(2n+1)!(2n+1)} \\
&= -\frac{1}{2} + \sum_{n=0}^{\infty} (-1)^n \frac{\left(\pi^2\right)^n}{(2n+1)!(2n+1)} \approx 0.08949
\end{aligned} \tag{3.9}
$$

Im integral gibt es kein N mehr somit ist der maximale Wert unabhängig von N.

4

4.a

$$
\begin{aligned}
\widehat{(\alpha f + \beta g)}(\omega) &= \int_{\mathbb{R}^n} (\alpha f + \beta g) e^{-i\omega t} dt \\
&= \alpha \int_{\mathbb{R}^n} f(t) e^{-i\omega t} dt + \beta \int_{\mathbb{R}^n} g(t) e^{-i\omega t} dt \\
&= \alpha \hat{f} + \beta \hat{g}
\end{aligned} \tag{4.1}
$$

4.b

$$\begin{aligned}\widehat{f(\alpha\,\cdot)}(k) &= \int_{\mathbb{R}^n} f(\alpha x)e^{ikx}dx \\ &= \int_{\mathbb{R}^n} \frac{1}{|\alpha|}f(x)e^{-ix\frac{k}{\alpha}}dx \\ &= \frac{1}{|\alpha|}\hat{f}(k/\alpha)\end{aligned} \tag{4.2}$$

4.c

$$\begin{aligned}\hat{f}_y &= \int_{\mathbb{R}^n} f(x)_y e^{-ikx}dx \\ &= \int_{\mathbb{R}^n} f(z)e^{-ik(z+y)}dz \\ &= \hat{f}e^{-iky}\end{aligned} \tag{4.3}$$

4.d

$$\begin{aligned}\int_{\mathbb{R}^n} \hat{f}(x)g(x)dx &= \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} f(y)g(x)e^{-ixy}dydx \\ &\overset{\text{fubini}}{=} \int_{\mathbb{R}^n} f(y)\hat{g}(y)dy \\ &= \int_{\mathbb{R}^n} f\hat{g}dx\end{aligned} \tag{4.4}$$

4.e

$$\begin{aligned}\overline{\hat{f}(k)} &= \overline{\int_{\mathbb{R}^n} f(x)e^{-ikx}dx} \\ &= \int_{\mathbb{R}^n} \overline{\hat{f}}(x)e^{-(-ikx)}dx \\ &= \hat{\overline{f}}(-k)\end{aligned} \tag{4.5}$$

4.f

$$\begin{aligned}\widehat{\partial_j f}(k) &= \int_{\mathbb{R}^n} \partial_j f(x)e^{-ikx}dx \\ &= \underbrace{[f(x)e^{-ikx}]_{\mathbb{R}^n}}_{=0} - \int_{\mathbb{R}^n} -ik_j f(x)e^{-ikx}dx \\ &= ik_j\hat{f}(k)\end{aligned} \tag{4.6}$$

Da die Funktion differenziebar mit kompakten Träger ist muss man die Integrationsgrenzen nur genug gross setzen und man erhält 0.

$$\begin{aligned}\partial_j\hat{f}(k) &= \frac{1}{\partial k_j}\hat{f}(k) \\ &= \int_{\mathbb{R}^n} \frac{1}{\partial k_j}\big(f(x)e^{-ikx}\big)dx \\ &= \widehat{-\imath x_j f}(k)\end{aligned} \tag{4.7}$$

5

5.a

$$f(x) = \chi_{[-\pi,\pi]}(x)\sin(x) \tag{5.1}$$

$$
\begin{aligned}
\hat{f}(\omega) &= \int_{-\infty}^{\infty} \chi_{[-\pi,\pi]}(x)\sin(x)e^{-i\omega x}dx \\
&= \int_{-\pi}^{\pi} \sin(x)e^{-i\omega x}dx \\
&= -\cos(x)e^{-i\omega x} + \int_{-\pi}^{\pi} -i\omega\cos(x)e^{-i\omega x}dx \\
&= -\cos(x)e^{-i\omega x} - i\omega\sin(x)e^{-i\omega x} + \int_{-\pi}^{\pi} \omega^2\sin(x)e^{-i\omega x}dx \\
&= \frac{1}{1-\omega^2}\left[-\cos(x)e^{-i\omega x} - i\omega\sin(x)e^{-i\omega x}\right]_{-\pi}^{\pi} \\
&= \frac{2i\sin(\pi\omega)}{\omega^2-1}
\end{aligned}
\tag{5.2}
$$

$|\omega| = 1$

$$
\begin{aligned}
\hat{f}(\pm 1) &= \int_{-\pi}^{\pi} \sin(x)e^{\mp ix}dx \\
&= \pm\frac{1}{2i}\int_{-\pi}^{\pi} e^{ix-ix} - e^{-2ix}dx \\
&= \pm\frac{1}{2i}\left[1 - \frac{1}{2i}e^{-2ix}\right]_{-\pi}^{\pi} = \mp i\pi + 1
\end{aligned}
\tag{5.3}
$$

5.b

$$
f(x) = e^{-|x|}\cos(x) \tag{5.4}
$$

$$
\begin{aligned}
\hat{f}(\omega) &= \int_{-\infty}^{\infty} \cos(x)e^{-(|x|+ix\omega)}dx \\
&= \int_{0}^{\infty} \cos(x)e^{-x(i\omega-1)}dx + \int_{0}^{\infty} \cos(x)e^{-x(i\omega+1)} \\
&= \sin(x)e^{-x(i\omega-1)} - \int_{0}^{\infty} -(i\omega-1)\sin(x)e^{-x(i\omega-1)}dx + \sin(x)e^{-(i\omega+1)} - \int_{0}^{\infty} -(i\omega+1)\sin(x)e^{-x(i\omega+1)}dx \\
&= \sin(x)e^{-x(i\omega-1)} - (i\omega-1)\cos(x)e^{-x(i\omega-1)} + \int_{0}^{\infty} (i\omega-1)^2\cos(x)e^{-x(i\omega-1)}dx \\
&+ \sin(x)e^{-x(i\omega+1)} - (i\omega+1)\cos(x)e^{-x(i\omega+1)} + \int_{0}^{\infty} (i\omega+1)^2\cos(x)e^{-x(i\omega+1)}dx \\
&= \frac{1}{1-(i\omega-1)^2}\left[\sin(x)e^{-x(i\omega-1)} - (i\omega-1)\cos(x)e^{-x(i\omega-1)}\right]_0^{\infty} + \frac{1}{1-(i\omega+1)^2}\left[\sin(x)e^{-x(i\omega+1)} - (i\omega+1)\cos(x)e^{-x(i\omega+1)}\right]_0^{\infty} \\
&= \frac{i\omega-1}{\omega^2+2i\omega} + \frac{i\omega+1}{\omega^2-2i\omega}
\end{aligned}
\tag{5.5}
$$

irgendwo ist ein Fehler. wahrscheinlihc ganz am Anfang :(

5.c

$$
f(x,y) = e^{-x^2-y^2-xy} \tag{5.6}
$$

$$
\begin{aligned}
\hat{f}(\omega) &= \int_{\mathbb{R}^2} e^{-x^2-y^2-xy-i(\omega_1 x+\omega_2 y)} dxdy \\
&= \int_{\mathbb{R}^2} e^{-(x^2+x(y+i\omega_1))} e^{-y^2-i\omega_2 y} dxdy \\
&= \int_{\mathbb{R}^2} e^{-\left(x+\frac{y+i\omega_1}{2}\right)^2+\frac{(y+i\omega_1)^2}{4}} e^{-y^2-i\omega_2 y} dxdy \\
&= \int_{-\infty}^{\infty} \sqrt{\pi} e^{-\frac{3}{4}y^2-iy(\omega_2-\frac{1}{2}\omega_1)-\frac{\omega_1^2}{4}} dy \\
&= \sqrt{\pi} e^{-\frac{\omega_1^2}{4}} \int_{-\infty}^{\infty} e^{-a(y+\frac{ib}{2a})^2-\frac{b^2}{4a}} dy, \qquad a=\frac{3}{4}, b=\omega_2-\frac{1}{2}\omega_1 \\
&= \pi\sqrt{\frac{4}{3}} e^{-\frac{\omega_1^2-\omega_1\omega_2+\omega_2^2}{3}}
\end{aligned} \tag{5.7}
$$

5.d

$$
f(x,y) = \begin{cases} 1 & 0 \le x \le y \le 1 \\ 0 & \text{sonst.} \end{cases} \tag{5.8}
$$

$$
\begin{aligned}
\hat{f}(\omega) &= \int_0^1 \int_x^1 e^{-i\omega_1 x - i\omega_2 y} dydx \\
&= \int_0^1 \left[\frac{1}{-i\omega_2} e^{-i(\omega_1 x+\omega_2 y)} \right]_x^1 dx \\
&= \frac{i}{\omega_2} \int_0^1 e^{-i(\omega_1 x+\omega_2)} - e^{-ix(\omega_1+\omega_2)} dx \\
&= \frac{i}{\omega_2} \left[\frac{ie^{-i\omega_2}}{\omega_1} e^{-i\omega_1 x} - \frac{i}{\omega_1+\omega_2} e^{-ix(\omega_1+\omega_2)} \right]_0^1 \\
&= e^{-i(\omega_1+\omega_2)} \left(\frac{1}{(\omega_1+\omega_2)\omega_2} - \frac{1}{\omega_1\omega_2} \right) + \frac{e^{-i\omega_2}}{\omega_1\omega_2} - \frac{1}{(\omega_1+\omega_2)\omega_2}
\end{aligned} \tag{5.9}
$$

$\omega_1 = 0, \omega_2 \neq 0$

$$
\begin{aligned}
\hat{f}(\omega) &= \int_0^1 \int_x^1 e^{-i\omega_2 y} dydx \\
&= \int_0^1 \left[\frac{i}{\omega_2} e^{-iy\omega_2} \right]_x^1 dx \\
&= \frac{i}{\omega_2} \int_0^1 e^{-i\omega_2} - e^{-i\omega_2 x} dx \\
&= \frac{i}{\omega_2} \left[xe^{-i\omega_2} + \frac{1}{i\omega_2} e^{-i\omega_2 x} \right]_0^1 = \frac{ie^{-i\omega_2}}{\omega_2} + \frac{1}{\omega_2^2} e^{-i\omega_2} - \frac{1}{\omega_2^2}
\end{aligned} \tag{5.10}
$$

$\omega_2 = 0 ...$
$\omega = 0$

$$
\begin{aligned}
\hat{f}(0) &= \int_0^1 \int_x^1 dxdy \\
&= \int_0^1 (1-x) dx \\
&= 1 - \frac{1}{2} = \frac{1}{2}
\end{aligned} \tag{5.11}
$$

6

6.a

$$\begin{aligned} ||f \star g||_1 &= \int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} f(x-y)g(y)dy \right| dx \\ &\leq \int_{\mathbb{R}^n} |f(x-y)|dx \int_{\mathbb{R}^n} |g(y)|dy \\ &= \int_{\mathbb{R}^n} |f(x)|dx \int_{\mathbb{R}^n} |g(y)|dy \\ &= ||f||_1 ||g||_1 \end{aligned} \tag{6.1}$$

6.b

man verwendet die ungelichheit von (6.1) damit ist es offensichtlihc da $f, g \in L^1(X)$

6.c

$$\begin{aligned} \partial((f \star g)(x)) &= \partial\left(\int_X f(x-y)g(y)dy \right) \\ &= \int_X \partial(f(x-y))g(y)dy + \int_X f(x-y) \underbrace{\partial(g(y))}_{=0} dy \\ &= ((\partial f) \star g)(x) \end{aligned} \tag{6.2}$$

6.d

die Eigenschaft in (6.2) sowieso, da das X nicht wichtig ist. Bei den anderen sit es auch klar ersichtlich da es periodisch wird. Wenn man die Substitution mache muss geht es immer noch da man über ganze Intervall integrieren muss.